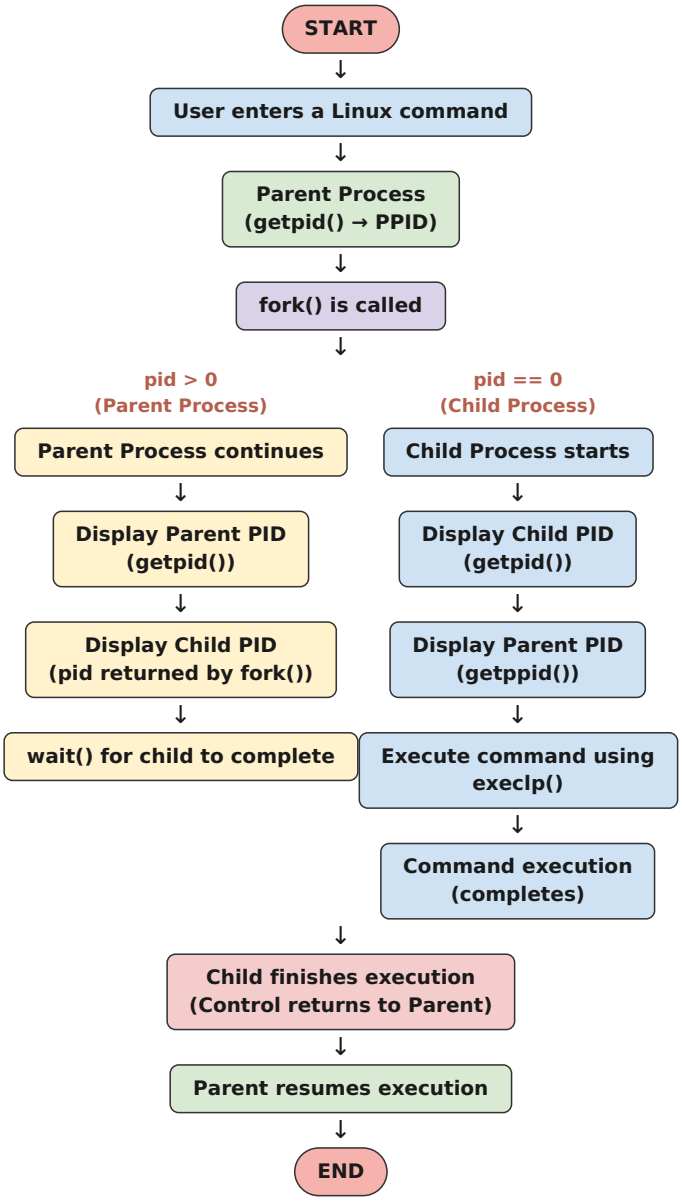
Practical Week -1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

a. Write a C code to create a process using fork() system call.
b. Read the user input as shell command.
c. Use a system call exec() to execute the shell command.
d. Use a wait() system call for waiting a parent process.
e. Display the PIDs of parent and child processes.



Example Output:

```
Enter a Linux command: ls
Parent Process
Child Process
Parent PID : 67
Child PID : 68
Child PID : 68
Parent PID : 67
Executing command: ls
```

```
CE1 FIRST.c File2.c Proces_States.c command_executor
filecopy.c hello.c
process_example session_4_a session_4_b.c
CE1.c File1.c First.c Process_States.c command_executor.c
fork_example.c.save my_shell process_example.c session_4_a.c
FIRST File1.txt Proces_States R1 filecopy hello
os-lab process_example.cyes session_4_b
Child process completed.
```

Sol:

gopinath@GOPINATH: ~ GNU nano 8.7.1 command_executor.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if (pid == 0)
    {
        // Child process

        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);

        perror("exec failed");
        exit(1);
    }

    else
    {
        // Parent process

        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
File1.txt        ecec_demo.c    fork_exec.c      task2.c
File2.txt        exec_demo      namedirectory    task3
OSSP             exec_demo.c    skill4           task3.c
OS_LAB           filename       task1
command_executor.c.save filename.c task1.c
ecec_demo        fork_exec      task2
gopinath@GOPINATH:~$ nano command_executor.c
gopinath@GOPINATH:~$ gcc command_executor.c -o command_executor
gopinath@GOPINATH:~$ ./command_executor
Enter a Linux command: ls
Parent Process
Parent PID : 550
Child Process
Child PID : 551
```

```
Parent PID : 550
Executing command: ls
File1.txt        ecec_demo.c    skill4
File2.txt        exec_demo      task1
OSSP             exec_demo.c    task1.c
OS_LAB           filename       task2
command_executor filename.c     task2.c
command_executor.c fork_exec    task3
command_executor.c.save fork_exec.c task3.c
ecec_demo        namedirectory
Child process completed.
gopinath@GOPINATH:~$ ./command_executor
Enter a Linux command: pwd
Parent Process
Parent PID : 617
Child Process
Child PID : 620
Parent PID : 617
Executing command: pwd
/home/gopinath
Child process completed.
gopinath@GOPINATH:~$
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Sol:

Command	What it shows
uname	Linux/system information
uname -a	Detailed kernel/system information
lscpu	CPU architecture and processor information
ps	Current processes
ps aux	Detailed process information
top	Real-time process and resource usage

```
gopinath@GOPINATH:~$ uname
Linux
gopinath@GOPINATH:~$ uname -a
Linux GOPINATH 6.18.33.2-microsoft-standard-WSL2 #1 SMP
PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
gopinath@GOPINATH:~$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
  Model name:                13th Gen Intel(R) Core(TM) i7
    CPU family:              6
    Model:                   186
    Thread(s) per core:      2
    Core(s) per socket:      6
    Socket(s):               1
    Stepping:                3
    BogoMIPS:                4991.99
    Flags:                   fpu vme de pse tsc msr pae mce
                             cx8 apic sep mtrr pge mca cmov pat
                             pse36 clflush mmx fxsr sse sse2 ss
                             syscall nx pdpe1gb rdtscp lm
                             constant_tsc rep_good nopl xtopology
                             nonstop_tsc cpuid tsc_known_freq pni
                             pclmulqdq vmx ssse3 fma cx16 pcid
                             sse4_1 sse4_2 x2apic movbe popcnt
                             tsc_deadline_timer aes avx f16c
                             rdrand hypervisor lahf_lm abm
                             3dnowprefetch ssbd ibrs ibpb stibp
                             ibrs_enhanced tpr_shadow vnmi ept
```

```
                                vpid ept_ad fsgsbase tsc_adjust
                                bmi1 avx2 smep bmi2 erms invpcid
                                rdseed adx smap clflushopt clwb
                                sha_ni xsaveopt xsavec xgetbv1 xsave
```

```
gopinath@GOPINATH:~$ ps
    PID TTY          TIME CMD
    319 pts/0    00:00:00 bash
    733 pts/0    00:00:00 ps
gopinath@GOPINATH:~$ ps aux
USER       PID %CPU %MEM    VSZ   RSS TTY  STAT START TIME COMMAND
gopinath     1  0.3  0.3  24328 15372 ?    Ss   05:20 0:03 /sbin/init
gopinath     2  0.0  0.0   3180  2208 hvc0 Sl+  05:20 0:00 /init
gopinath     7  0.0  0.0   3180  2048 hvc0 Sl+  05:20 0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --
pipe-f
gopinath    53  0.0  0.4  50396 16804 ?    S<s  05:20 0:00 /usr/lib/systemd/systemd-journald
systemd+    70  0.0  0.3  22416 14468 ?    Ss   05:20 0:00 /usr/lib/systemd/systemd-resolved
gopinath    92  0.1  0.3  35140 12256 ?    Ss   05:20 0:01 /usr/lib/systemd/systemd-udevd
gopinath   153  0.0  0.0   2888  1948 ?    Ss   05:20 0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F
1
gopinath   154  0.0  0.0   4472  3100 ?    Ss   05:20 0:00 /usr/sbin/cron -f -P
message+   155  0.0  0.1   8820  5316 ?    Ss   05:20 0:00 @dbus-daemon --system --address=systemd: --nofork --
nopidfile
gopinath   159  0.0  0.7  44096 29712 ?    Ss   05:20 0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-
startup-t
gopinath   168  0.0  0.2  18436  9388 ?    Ss   05:20 0:00 /usr/lib/systemd/systemd-logind
syslog     175  0.0  0.1 220548  5568 ?    Ssl  05:20 0:00 /usr/sbin/rsyslogd -n -iNONE
gopinath   216  0.0  0.0   5492  2660 tty1 Ss+  05:20 0:00 /sbin/agetty --noreset --noclear --issue-file=/etc/issue:
gopinath   225  0.0  0.8 123460 32772 ?    Ssl  05:20 0:00 /usr/bin/python3 /usr/share/unattended-
upgrades/unattended-up
_chrony    226  0.0  0.2  23588 11084 ?    S    05:20 0:00 /usr/sbin/chronyd -n -F 1 -x
```

```
gopinath@GOPINATH:~$ top
top - 05:43:42 up 2:51,  1 user,  load average: 0.09, 0.04, 0.
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem :   3758.6 total,   3048.4 free,    473.1 used,    312.
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3285.

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
    1 gopinath  20   0   24328  15372  11568 S   0.0   0.4
    2 gopinath  20   0    3180   2208   2072 S   0.0   0.1
    7 gopinath  20   0    3180   2048   1940 S   0.0   0.1
   53 gopinath  19  -1   50396  16804  15500 S   0.0   0.4
   70 systemd+  20   0   22416  14468  11972 S   0.0   0.4
   92 gopinath  20   0   35140  12256   9184 S   0.0   0.3
  153 gopinath  20   0    2888   1948   1820 S   0.0   0.1
  154 gopinath  20   0    4472   3100   2852 S   0.0   0.1
  155 message+  20   0    8820   5316   4636 S   0.0   0.1
  159 gopinath  20   0   44096  29712  14576 S   0.0   0.8
  168 gopinath  20   0   18436   9388   8144 S   0.0   0.2
  175 syslog    20   0  220548   5568   4648 S   0.0   0.1
  216 gopinath  20   0    5492   2660   2476 S   0.0   0.1
  225 gopinath  20   0  123460  32772  18164 S   0.0   0.9
  226 _chrony   20   0   23588  11084   7128 S   0.0   0.3
  237 _chrony   20   0   12096   2452   1808 S   0.0   0.1
  316 gopinath  20   0    3184   1112    980 S   0.0   0.0
  317 gopinath  20   0    3200   1248   1108 S   0.0   0.0
  319 arikela   20   0    6268   5532   3864 S   0.0   0.1
  322 gopinath  20   0    8648   5544   4700 S   0.0   0.1
  385 arikela   20   0   22340  13604  10976 S   0.0   0.4
  387 arikela   20   0   22920   4092   2088 S   0.0   0.1
  412 arikela   20   0    6136   5424   3864 S   0.0   0.1
  755 arikela   20   0    8164   6064   3948 R   0.0   0.2
```